Comparing Alternatives for Managing Hydrometeorological Financial Risk for Hydropower Suppliers

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Abstract: Droughts and heatwaves create significant financial risk for hydropower suppliers, as drought reduces generation and revenues, while heatwaves increase demand and often costs. While many suppliers manage their risk independently, some are government-supported. This leads to an important question: How do the costs of a government-supported risk management strategy compare with a self-supported strategy? This research provides a framework to address this question, applying it to a hydropower supplier in the United States with access to a government-supported line of credit, the Bonneville Power Administration (BPA). The analysis includes models to stochastically simulate weather, electricity prices, and the BPA's operations. Results suggest that government support effectively mitigates the BPA's hydrometeorological financial risk but at costs to taxpayers that can exceed $1 billion over 20 years. Replacing the line of credit with index insurance comparably manages financial risk at lower costs limited to the BPA and its customers. This framework for comparing government- and self-supported approaches may prove useful to hydropower suppliers and governments, as they decide how to manage hydrometeorological financial risk, especially in lower-income countries with limited public funds. **DOI: 10.1061/JWRMD5.WRENG-6573.**

Introduction

In recent years, severe droughts around the world have dramatically reduced hydropower generation and impacted regional electricity supplies (Hansrood 2022; Bloomberg News 2022). In 2022 alone, countries as far ranging as China, the United States (US), and Zimbabwe experienced droughts that led to decreases in hydropower supplies and, correspondingly, revenues. These decreases were difficult to manage given that suppliers often have high fixed costs, and so revenue reductions lead to budget shortfalls. In addition, many hydropower suppliers hold firm electricity delivery contracts. During droughts, they may be forced to purchase compensatory electricity on the wholesale market to meet these obligations, increasing their costs. The impact of drought can be exacerbated by heatwaves, which lead to spikes in electricity demand alongside drought-related decreases in supply, contributing to increases in electricity prices. This combination of weather-related impacts on demand, supply, and electricity prices led to multimillion dollar losses for hydropower suppliers in 2022 (Franke 2022; Canale et al. 2019; Fick 2022). Nonetheless, while significant attention has been given to managing the supply risk arising from extreme weather (WMO 2022), less attention has been given to the emerging challenge of managing the accompanying financial risk, i.e., volatility in financial outcomes, such as revenues, costs, and the metric of primary focus in this analysis, i.e., net revenues.

Managing this weather-related financial risk is critical to hydropower suppliers and utilities with significant hydropower assets, whose largest costs are often associated with borrowing for capital expenditures. In the worst case, a supplier may default on its debt obligations, but even an increase in the risk of this occurring can be costly in terms of raising borrowing costs. Credit rating agencies (e.g., Moody's, S&P Global, Fitch) evaluate the creditworthiness of public and private entities, and a downgrade based on changes in financial risk increases the cost of future borrowing via higher interest rates (Lowrey 2017). And over the past decade, each of the major agencies have issued statements describing the financial consequences of droughts for hydropower (Soares and Hess 2014; Cregger et al. 2021; Rack 2020; Leong et al. 2022; Kim and Krummenacker 2020).

While several traditional risk management tools such as reserve funds and lines of credit are commonplace, these are often less cost effective for managing extreme events (Mechler et al. 2010; The World Bank 2017). Consequently, there has been growing attention on the importance of risk transfer tools, which can more cost-effectively manage low-probability but high-impact events (CNA Insurance 2016; Kron et al. 2019). For sectors in which there is a clear relationship between hydrometeorological conditions and outcomes, such as hydropower, index-based (parametric) instruments can offer a lower cost option than traditional, indemnity-based insurance, which requires individual and often subjective loss assessments. Given their advantages, index-based instruments have been

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the subject of extensive research across sectors, including agriculture (Barnett and Mahul 2007), water supply (Zeff and Characklis 2013; Baum et al. 2018), electric power (Hamilton et al. 2020; Meyer et al. 2017; Kern et al. 2015; Foster et al. 2015; Liao et al. 2021; Matsumoto and Yamada 2021), inland navigation (Meyer et al. 2016, 2020), and biofuel production (Kleiman et al. 2022). Index-based instruments have also already been marketed in a number of settings related to natural hazard risks (IIF, n.d.; USDA Risk Management Agency, n.d.; Evans 2019; Greatrex et al. 2015; The World Bank 2013; Swiss Re 2018, 2022). A recent example involves an index insurance contract taken out by the hydropower-dependent Uruguayan electric utility UTE, underwritten by the World Bank in partnership with reinsurers Swiss Re and Allianz (Swiss Re 2018; The World Bank 2013). This contract used a rainfall-based index that triggered payouts to UTE under predefined drought conditions in order to compensate it for the costs of purchasing oil to run expensive thermal generation units replacing lost hydropower. A primary benefit of such a contract is its fixed cost, which stands in contrast to tools like reserve funds and lines of credit, whose costs can be highly variable, although there is no research comparing which parties shoulder the costs of these different tools.

Another solution, however, is for the public sector to assume responsibility for weather-related financial risk. This was the approach taken in France in July of 2022 when the French government announced its intent to fully nationalize Électricité de France (EDF), a decision driven in part by the financial impacts of severe droughts and heatwaves across continental Europe (De Beaupuy et al. 2022). In so doing, the government accepted the risk associated with volatility in supply and demand, with the intent of stabilizing electricity prices, ultimately transferring the costs of risk management to taxpayers (Lucas 2014; Klein 1997). The contrast between the approaches taken in Uruguay and France raises an important and unanswered question: Does government support provide more effective risk management than index insurance? This work addresses this question by contrasting government-supported and independent risk management strategies for a hydropower supplier. Evaluating each strategy's effectiveness in mitigating severe losses along with the costs to consumers, the supplier, and the government can inform public sector involvement.

One entity that has access to government-supported and independent risk management tools, and that might benefit from index insurance, is the Bonneville Power Administration (BPA). Located in the US Pacific Northwest (PNW), the BPA was established to market and deliver electricity from the 31 federal dams in the Federal Columbia River Power System (FCRPS) (Fig. 1) that contribute an average of 50% of the region's generation (NWPCC 2019). This hydropower dependence makes the BPA financially vulnerable to hydrologic variability (Denaro et al. 2022; Kim and Krummenacker 2020; Kim and Sabatelle 2022; Kim et al. 2023), a trait major credit rating agencies have been noting since 2012 (Kim and Sabatelle 2012).

To address its financial vulnerability, the BPA has developed a three-pronged risk management strategy. It first draws on a reserve fund, turning to a line of credit as potential losses mount and then triggering tariff surcharges as revenue shortfalls occur. The line of credit is of particular note: though the BPA's mandate is to operate as a financially independent organization, it has access to a line of credit with the US treasury, which has been depleted and expanded several times since its inception in 1974. For credit rating agencies, the expectation of continued government support reduces concerns over the BPA's finances (Kim and Krummenacker 2019; Kim and Sabatelle 2022). However, the line of credit was on track to be depleted by 2023 (BPA 2018), contributing to Moody's downgrade of the BPA's credit rating in 2020. Moody's report cited concerns over the BPA's diminishing ability to manage its hydrologic risk and

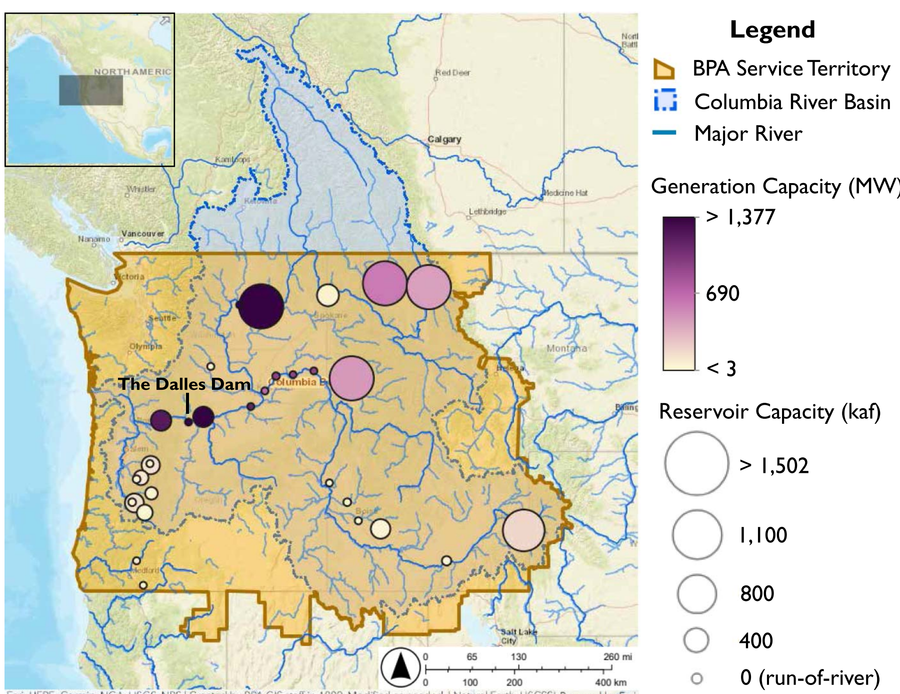

Fig. 1. Map of Bonneville Power Administration service area, the 219,500 mi^2 Columbia River Basin, and locations of the 31 dams managed by BPA. (Map from Esri, HERE, Garmin, NGA, USGS, NPS; Created by BPA GIS staff in 1989. Modified as needed. Natural Earth, USCSSI Powered by Esri.)

declining access to the line of credit (Kim and Krummenacker 2020), factors that have been noted by other rating agencies (Bodek and Panger 2021). The government, for its part, also appears to have tacitly acknowledged the BPA's exposure and expanded the line of credit as part of the 2021 Infrastructure Investment and Jobs Act, mitigating rating agencies' concerns, though the BPA's continued financial vulnerability contributed to a change in rating outlook from stable to negative in 2023 (Kim et al. 2023). This intervention has parallels to the French government's actions to manage the EDF's hydrometeorological financial risk. Just as for the EDF, questions arise as to whether the BPA could independently manage its hydrometeorological financial risk, here characterized in terms of its variable net revenues, as opposed to relying on government support. Both approaches involve the BPA managing its risk, but the costs and groups to which the costs accrue differ.

Given the BPA's financial challenges and access to multiple risk management tools, this work aims to contrast the performance of government support versus independent risk management, including use of a new index insurance contract. The exploration of this question in the context of the BPA is facilitated by a series of linked models that represent the natural (temperature, streamflow), engineered (reservoir operations), and economic/financial (electricity markets, the BPA's business operations) elements of the system. It builds on earlier work by Denaro et al. (2022), who assessed the BPA's financial vulnerability prior to the federal government's intervention. And, though this dramatic increase in government support via the expanded line of credit forms the primary motivation for this analysis, the question is of greater significance for the many utilities and hydropower suppliers facing similar hydrometeorological risks but which lack the BPA's sophisticated, multilayered risk management strategy. For these utilities, the methods for disentangling the costs of risk management and the potential for designing new, self-supported strategies can be used to evaluate and prioritize use of different risk management tools. This question may be particularly relevant for the growing number of hydropower suppliers and the governments in countries where they reside as they seek to manage the rising frequency of coincident droughts and heatwaves (Mazdiyasni and AghaKouchak 2015; Alizadeh et al. 2020).

Methods

The modeling approach involves jointly generating synthetic conditions related to hydrometeorology, electricity supply, and electricity demand, which are linked to a model of the BPA's operations (Fig. 2). In earlier work by Denaro et al. (2022), this modeling platform was used to assess the BPA's hydrometeorological financial risk and the effectiveness of its current risk management strategy, which was found to leave it exposed to large losses. In this research, that platform has been adapted to compare the effectiveness of the BPA's current risk management strategy, which relies on the government-supported line of credit, with a self-supported risk management strategy in which the BPA uses index insurance in place of the line of credit.

Weather and price data are generated via the California and West Coast Power System model (CAPOW) (Su et al. 2020a), composed of two components. The first is a stochastic weather generator that simulates daily temperature, wind, irradiance, and streamflow within the Mid-Columbia (Mid-C) and California Independent System Operator (CAISO) electricity markets. The generator simulates weather conditions in a manner that is representative of historical conditions but synthetically expands beyond the limited available observations so as to capture extreme events outside the historical record. Weather conditions are then fed into a unit-commitment/economic dispatch (UC/ED) model, which is used to simulate regional generation and electricity prices within the Mid-C and CAISO using synthetic hydrometeorological data as inputs. Electricity prices and hydropower operations are then used as inputs to a model of the BPA's business operations that translates electricity supply and demand into annual net revenues. Model results are used to evaluate the effectiveness and costs of financial risk management strategies. More details on each component of the approach are provided in the following sections.

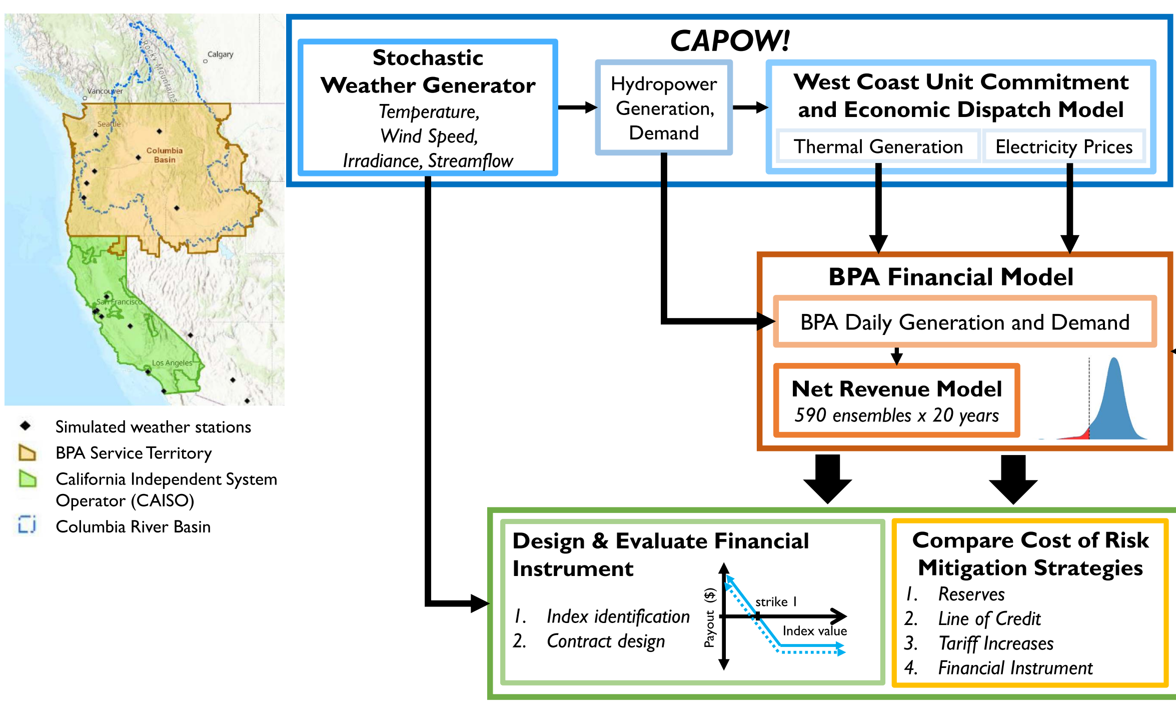

Fig. 2. Integrated modeling framework.

California and West Coast Power System Model

The CAPOW model is a fundamental component of the analytical framework. This section briefly describes the model [for a comprehensive description, see Su et al. (2020a)].

Stochastic Weather Generator

The stochastic weather generator uses coincident data on temperature, wind, and irradiance at 17 sites along the West Coast, jointly available between 1998 and 2017, from the Global Historical Climatology Network (Menne et al. 2012) and the National Solar Radiation Database (Sengupta et al. 2018). Historical data are used to generate synthetic daily weather conditions that influence electricity supply and demand. Seasonal patterns are removed from observations to obtain detrended residuals that are mean-shifted and log-transformed, a process that results in a distribution that is approximately Gaussian. A vector autoregressive model is used to generate cross-correlated, whitened residuals for each weather variable. Seasonal patterns are added to the residuals once they are backtransformed, producing a final synthetic value. Streamflow is synthetically generated at an annual timestep using a Gaussian copula linked to temperature and disaggregated to the daily level using a nearest-neighbor clustering approach. The final synthetic data set, which accurately reproduces historical patterns, including spatial, temporal, auto, and cross-correlations, allows for the characterization of hydrometeorological variability in the BPA's service area.

Unit Commitment/Economic Dispatch Model

The stochastic weather generator provides the inputs necessary for simulating electricity demand and supply. With regard to the former, the strong correlation with hydrometeorological conditions (i.e., temperature, wind speed) allows for the simulation of demand via multivariate regression ($r^2 = 0.73$ at an hourly time-step).

With respect to supply, hydropower generation is simulated based on streamflow and hourly demand. Dams accounting for more than 85% of hydropower capacity in the PNW are mechanistically modeled (i.e., via a water balance approach) using the US Army Corps of Engineers' Hydro System Seasonal Regulation model (HYSSR) for the Federal Columbia River Power System and ResSim for dams on the Willamette River, the only major tributary to the Columbia not included in HYSSR. Modeled dams account for 87.5% of the BPA's hydropower capacity [see Su et al. (2020a), supplemental information]. Smaller dams accounting for the remaining 15% of capacity are statistically modeled as generating hydropower in proportion to nearby dams according to historical relationships. Historical simulations show strong agreement for the BPA's hydropower generation at the monthly timestep ($r^2 = 0.76$) (Denaro et al. 2022). Thermal generators (e.g., natural gas, coal) are dispatched in the UC/ED model according to marginal cost, after lower marginal cost sources of electricity (i.e., solar, wind, and hydropower) have been exhausted. The cost of the most expensive generator used to meet demand sets the wholesale electricity price. Interconnections between the Mid-C and CAISO markets are included, using historical patterns and transmission capacity to statistically simulate imports and exports. For the purposes of isolating the impact of hydrometeorological variability on the electricity market, natural gas prices are held constant, as in Su et al. (2020b). This suggests that results may underestimate the range of volatility in the BPA's annual net revenues and thereby the full extent of its financial risk.

Our analysis uses the 1,180 simulated years from CAPOW also used by Denaro et al. (2022) and Su et al. (2020a). Simulated hydrometeorological conditions and electricity prices are used as inputs to a financial model for the BPA, described in the following section, to evaluate the range of net revenues (financial risk) under extreme hydrometeorological conditions. Because historical hydrometereological conditions do not exhibit significant autocorrelation [see Denaro et al. (2022) and supplemental materials (SM)], these runs were shuffled (i.e., randomly sampled without replacement) 10 times to provide 11,800 years. These 11,800 years were subset into 590 unique 20-year ensembles to capture the temporal dynamics of financial risk. The shuffled ensembles capture the variable capacity of existing financial tools to absorb hydrometeorological financial risk over sequential years.

Bonneville Power Administration Financial Model

The BPA, which in 2022 had gross sales of \$4.6 billion and operating expenses of \$3.4 billion (BPA 2022), has a service area spanning the US PNW. It is subject to two compounding hydrometeorological financial risks. The first, of concern for any power utility, is the impact of temperature on demand (Apadula et al. 2012; Yao 2021). Heatwaves can drive sudden spikes in demand that drive up wholesale electricity prices, as more expensive thermal generation is activated. The second is the impact of hydrology on electricity supply in hydropower-dominated systems such as the Mid-C market, within which the BPA operates, which has contributed to swings of over \$1 billion in the BPA's annual net revenues (Kim et al. 2016).

Long-term contracts the BPA holds with regional customers ("preference customers") to provide fixed quantities of electricity at a set price define the BPA's firm demand and baseline generation targets, irrespective of temperature fluctuations. Surplus generation is sold on the wholesale Mid-C market or exported into the neighboring CAISO market. Yet, should the BPA's generation fall short of its fixed obligations to preference customers, as during a drought, it must still meet its contractual obligations and is therefore forced to purchase additional electricity on the wholesale market, leading to higher costs and increased budgetary losses. When droughts coincide with heatwaves and demand spikes, wholesale electricity prices are driven even higher than during droughts alone, generating the greatest level of losses for the BPA and similar utilities.

The majority of the BPA's hydrometeorological financial risk is mitigated using three tools. First, the BPA has a reserve fund that it aims to keep at between 60 and 120 days' worth of operating expenses (\$304 million–\$608 million) (BPA 2017). When reserves are insufficient to offset losses, the BPA can turn to its line of credit with the US treasury (i.e., the borrowing authority), which is also used to support long-term capital expenditures. Finally, tariff surcharges up to \$5/MWh [on top of the 2001–2022 average preference customer price of \$41.88 (US EIA 2023)] are triggered via the cost recovery adjustment clause (CRAC) when there are net revenue shortfalls of over \$5 million (BPA 2019).

The BPA business operations model is linked to CAPOW and simulates revenues, costs, and net revenues. Model results show that it is able replicate historical net revenues accurately ($r^2 = 0.84$). The findings from Denaro et al. (2022) indicate that the BPA is exposed to uncovered losses of over \$200 million in 5% of years (using 2018 values as initial conditions), even after fully deploying its risk management strategy, thus suggesting a need FOR additional risk management tools to reduce uncovered losses to a value the BPA could reasonably be expected to offset in a subsequent year (average annual net revenues per Denaro et al.: \$136 million).

The model used by Denaro et al. (2022) is the foundation for this analysis, but it has been modified to include more detailed consideration of each risk management tool and to incorporate payments made to the line of credit and reserves in calculations for net revenues. A second change is limiting the model to focus exclusively on operational and short-term costs, i.e., excluding capital expenditures and use of the line of credit for that purpose. As such, annual access to the line of credit is limited to the liquid \$750M

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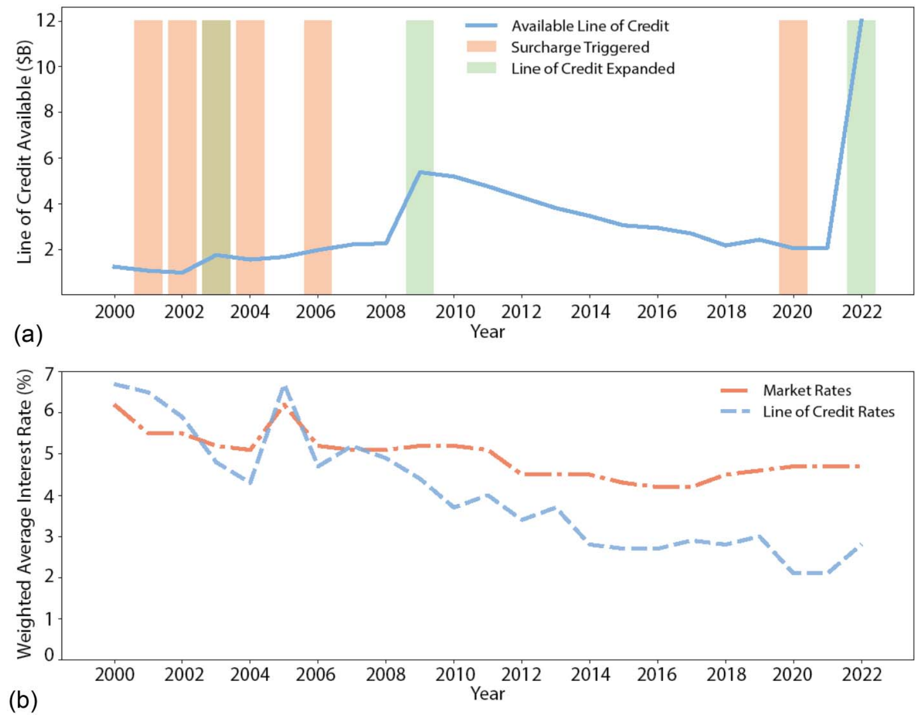

Fig. 3. Timeline describing the Bonneville Power Administration's line of credit with the US government: (a) available line of credit and associated periods with tariff surcharges triggered by net revenue shortfalls; and (b) difference between the weighted average interest rate for the BPA's federal (line of credit) debt and market (nonfederal) sources.

Treasury Facility, which is available on a two-year basis. This isolates the impacts of sudden hydrometeorological extremes on short-term financial risk. Starting conditions for each 20-year ensemble have also been updated as the average of historical (2018–2022) available reserves, operating costs, and electricity prices (see Table S1 in SM).

Risk Management Strategies

The purpose of this paper is to compare the costs and advantages of government-supported risk management, using the line of credit with reserves and tariff surcharges, with self-supported risk management strategies using the reserve fund and tariff surcharges in addition to a new index-based instrument. For the purposes of providing a baseline understanding of the BPA's financial risk, outcomes are initially presented assuming that no risk management tools are deployed (Strategy 0). Strategy 1 (reserves + surcharges) takes into account the BPA's existing self-supported tools, namely, cash reserves and tariff surcharges. It reflects the culmination of the BPA's trajectory prior to 2021, that is, depletion of the line of credit [as seen in Fig. 3(a)] by 2023 (BPA 2018) and access only to the reserves and surcharges.

Strategy 2 includes the BPA's government-supported tool, the line of credit, and reflects the willingness of the government to expand it, as occurred in the 2021 Infrastructure Investment and Jobs Act (DeFazio 2021). The expansion raised the cap on the line of credit to \$17.7 billion, providing ample borrowing capacity to insulate the BPA from hydrometeorological financial risks over the medium-term. The line of credit, and the government's willingness to expand it, is modeled by increasing the cap on the line of credit each time it falls below \$1.5 billion, the BPA's minimum target (BPA 2022). Historically, the BPA borrows more from the line of credit each year than it repays, suggesting that the government long-term absorbs some losses or foregone repayments. To provide an upper and lower bound for these foregone repayments, two repayment scenarios are modeled, one in which no payments are made, and another in which the BPA dedicates 100% of its positive net revenues in each year to meeting scheduled annual debt service (Strategies 2a, unlimited line of credit—no repayment; and 2b, unlimited line of credit—full repayment, respectively). Strategy 3 provides a possibility for managing the BPA's financial risk without access to the line of credit: supplementing its reserve fund and tariff surcharges with a new financial instrument (index insurance).

Table 1 summarizes the three strategies evaluated, selected to contrast independent risk management (Strategies 1 and 3) versus government-supported risk management (Strategies 2a and 2b). Strategy 0 has no risk management tools in place.

Evaluating the Costs of Government Support

The costs of the low interest line of credit supported by the US treasury are largely borne by the US government (i.e., taxpayers). The costs are direct and indirect, but can be split into:

1. Opportunity cost: the cost to the US government of holding unused, liquid funds for the BPA's use (i.e., the \$750 million Treasury Facility).
2. Foregone repayments: the difference in the present value (PV) of payments that the US government should have received from the BPA to fully repay the annual amortized value of borrowing on the line of credit and the simulated level of repayment by the BPA.
3. Interest rate subsidy: the difference between the cost of borrowing at federally subsidized and market-based (nonfederal) interest rates for the BPA (i.e., the difference in interest rates).

For the purpose of this work, the sum of these three costs is considered the government's total cost of providing the line of credit to the BPA. Each cost is described in greater detail in the following.

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Table 1. Financial strategies and associated line of credit and reserve caps

Name	Reserves	Line of credit	Repayment schedule	Index insurance
0. No risk management	\$0	\$0	—	No
1. Reserves + surcharges	\$608 million	\$0	—	No
2. Unlimited line of credit	\$608 million	Expanded as necessary	—	No
2a. No repayment	—	—	None	—
2b. Full repayment	—	—	Maximum[a]	—
3. Index insurance	\$608 million	\$0	—	Yes

Note: Strategies 1 (no line of credit) and 3 (index insurance) reflect risk management independent of government intervention.
[a]Maximum repayment is the use of up to 100% of positive net revenues in any single year to meet annual scheduled repayment obligations.

The opportunity cost of holding liquid funds for the BPA's access is estimated based on ways to otherwise invest or spend these funds. There is no single way to estimate the value of an alternative investment for a national government, for whom an investment in infrastructure may yield varying monetary (and nonmonetary) returns, but one way to quantify annual opportunity cost (O_c) is the potential interest earned from investing those funds. In this case, the investment is the unused portion of the line of credit, represented by the \$750 million per year that the BPA has "overnight" access to less the amount it has borrowed (the cost of the latter is described below). This unused portion is considered to earn no return, as it is allocated for the BPA's use. The opportunity cost is calculated for each year within the 20-year ensemble and then discounted back to PV

$$O_c = \sum_{t=1}^{20} \frac{(TF_t \cdot (1+i)^{t_1}) - TF_t}{(1+d)^t} \tag{1}$$

where TF_t = the value of the unused portion of the line of credit (the \$750 million Treasury Facility) in year t within the ensemble; i = the interest rate, expressed as a decimal value; and t_1 = the time horizon for the investment (one year). The discount rate (d) is set to 2.5%, consistent with guidelines from the US Office of Management and Budget (OMB 2022), and the risk-free alternative investment return (i) is 3.74%, the 2000–2022 average interest rate for 20-year US treasury bonds (Board of Governors of the Federal Reserve System 2022).

The cost of the portion of the line of credit that the BPA borrows is valued as a function of the expected repayment (principal plus interest). The government has never formally forgiven the BPA's debt, but the line of credit has historically been drawn down faster than it is repaid [Fig. 3(a)], and the BPA has the ability to indefinitely defer treasury payments with limited consequences from the treasury (Kim et al. 2023). Furthermore, over longer-term time horizons, the government has demonstrated willingness to raise the amount of funding available via the line of credit each time it approaches depletion, as in 2021 [Fig. 3(a)], effectively turning this loan into a partial grant.

The expected repayment is calculated by amortizing the amount of the line of credit used in a given year over a 50 year time horizon, the repayment schedule typically used by the BPA (BPA 2022). Missed or incomplete payments contribute to foregone repayments. Strategies 2a and 2b offer an upper and lower bound for foregone repayments, \$0 repayment, or maximum possible repayment. The former is equivalent to debt forgiveness, with the government forgiving any borrowing against the line of credit, while the latter represents a situation in which the BPA dedicates all positive net revenues in any year to repaying the line of credit. Maximum repayment is thus calculated by setting the BPA's repayment of the line of credit debt as the minimum of the full value due according to the amortization schedule or 100% of positive net revenues in a given year. The cost of the foregone repayments is then the difference between the amount due to fully repay the line of credit (i.e., the payment according to the amortization schedule) and the amount that is repaid according to the two repayment strategies. It should be noted that the BPA incurs costs that are the inverse of the government's; using Strategy 2a, the BPA has no repayment costs \$0. On the other hand, the BPA's costs for Strategy 2b are the discounted repayments. Costs are discounted over 20 years within each ensemble using a discount rate (d) of 2.5%. Annual foregone repayments (fr) are summed to calculate the PV, such that

$$fr = \sum_{t=1}^{20} \left[p_t \cdot \frac{i(1+i)^{50}}{(1+i)^{50} - 1} - \text{paid}_t \right] \tag{2}$$

$$\text{Cumulative } PV = \sum_{t=1}^{20} \frac{fr}{(1+d)^t} \tag{3}$$

where t = the ensemble year; paid_t = the modeled repayment; and p_t = the amount of the line of credit used each year (i.e., principal owed).

The third cost is the subsidy implicit in the line of credit's below market interest rate. This cost can be estimated as the difference in the interest rates charged for use of the line of credit versus those charged for borrowing from nongovernmental sources (i.e., the market). Between 2000 and 2022, the difference in these weighted average interest rates (WAIs) was as high as 3% [see Fig. 3(b)]. We thus calculate the difference in returns (i.e., interest payments) for the US treasury from simulated lending to the BPA via the line of credit using average historical market interest rates versus average historical Treasury interest rates, discounting the values of each over the 20-year ensembles.

Financial Instrument

An index-based financial instrument is designed for use in conjunction with the BPA's reserve fund and tariff surcharges. This insurance, paid for by the BPA, would allow it to manage its financial risk independently of government support. Like indemnity-based insurance, index insurance includes a fixed payment (premium) to the insurer in exchange for a payout under specified conditions. In the case of an index-based contract, the conditions for payouts are defined by an index that is a function of some readily measurable parameter or set of parameters (e.g., streamflow, temperature) correlated with financial losses. Premiums are often calculated as the sum of expected payouts and "loading," with the latter compensating the insurer for accepting risk on behalf of a client and administrative expenses.

The major challenge in formulating index insurance contracts is identifying an index that correlates well with losses. A high level of correlation suggests the instrument has low basis risk, i.e., provides payouts whose timing and magnitudes correspond well to losses. Moreover, indices should be readily interpretable and not be easily manipulable by either the buyer or seller. In this work, an index-based

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instrument is designed such that the strategy achieves the same 95% value at risk (VaR), the value below which the worst 5% of outcomes fall, as strategies relying on reserves, surcharges, and the line of credit, facilitating comparisons between them.

The index is based on the relationship between weather conditions and net revenues and is incorporated into a risk management strategy along with reserves and tariff surcharges. Development of an effective index, however, is hindered by the impact of tariff surcharges on the relationship between hydrometeorology and net revenues, as the surcharges increase the revenues per unit of electricity sold over the course of each 20-year ensemble, meaning that the same levels of streamflow and electricity generation yield greater net revenues in later years of the ensembles. To address this issue, the index is designed without consideration of tariff surcharges (i.e., only reserves). A clear link emerges between net revenues and streamflow at the Dalles Dam (TDA), one of the most downstream dams in the Columbia River Basin, and therefore one that accounts for aggregated flows from most points within the basin. This relationship is consistent with the clear link between streamflow and electricity generation (Karakoyun 2024), which is a critical factor in net revenues. The Dalles flows are simulated as adjusted routed flows (ARF), which reflect the amount of streamflow at the Dalles were there no upstream dams (Dakhlalla et al. 2020) and thus better reflect overall water availability instead of upstream dam releases. However, the r^2, observed between flow at the Dalles Dam and net revenues, is only 0.61, implying substantial basis risk. Lack of fit is particularly evident in drier years [see Fig. 4(a)], when the r^2 falls to 0.54, suggesting that flow at the Dalles alone may not be an effective proxy for net revenues.

Incorporation of average annual Mid-C electricity prices (\$/MWh) improves the index's correlation with net revenues. Prices are an indicator of regional supply and demand and communicate information valuable in assessing the cost to the BPA of purchasing compensatory electricity during drought. Electricity price is also attractive as an index parameter, as it could not reasonably be manipulated by the BPA given that its role is simply to market the power generated under conditions dictated by the US Army Corps of Engineers and the US Bureau of Reclamation.

Once Mid-C prices are considered, the index (v) representing net revenues is defined as

$$v(\$) = (5.1 \times 10^8) + (708 \cdot \mathrm{TDA}) - (2.6 \times 10^7 \cdot \text{Mid-C}) \quad (4)$$

This index captures the positive impact of streamflows, which lead to higher generation, and the negative impact of high electricity prices during supply shortfalls. There is a strong correlation between this index and annual net revenues ($r^2 = 0.75$) across the full synthetic data set [see Fig. 4(b)] and below the strike ($r^2 = 0.87$), a substantially better than the fit with Dalles flow alone. Star points in the figure, representing historical data from 2009 to 2018, provide some measure of validation that the index captures net revenue variability, particularly in years with negative net revenues. The historical data are limited because (1) the advent of hydraulic fracturing in 2009 dramatically changed the electricity price regime (Thomson Reuters 2023; US EIA 2023), and (2) ARF streamflow data beyond 2018 was not available at time of submission. The comparison highlights the improved predictive ability of the composite index (vis-à-vis the streamflow only index) for years with negative net revenues. Years of poor index performance, namely, 2014, 2015, and 2018, did not experience the combination of low streamflows (generation) and high prices (demand) that the index was designed to capture and instead experienced higher flows and prices (i.e., 2014 and 2018) or lower flows and low prices (i.e., 2015). See SM for further details.

The index insurance contract is defined by its strike (S), the threshold below which payouts occur, and the payout function, which describes how payouts increase as the index value declines below the strike. In this case, the strike is set at the fifth percentile of TDA flow, 133,907 cu ft per second (cfs), a value that the BPA is unable to manipulate to trigger payouts. The contract provides payouts equal

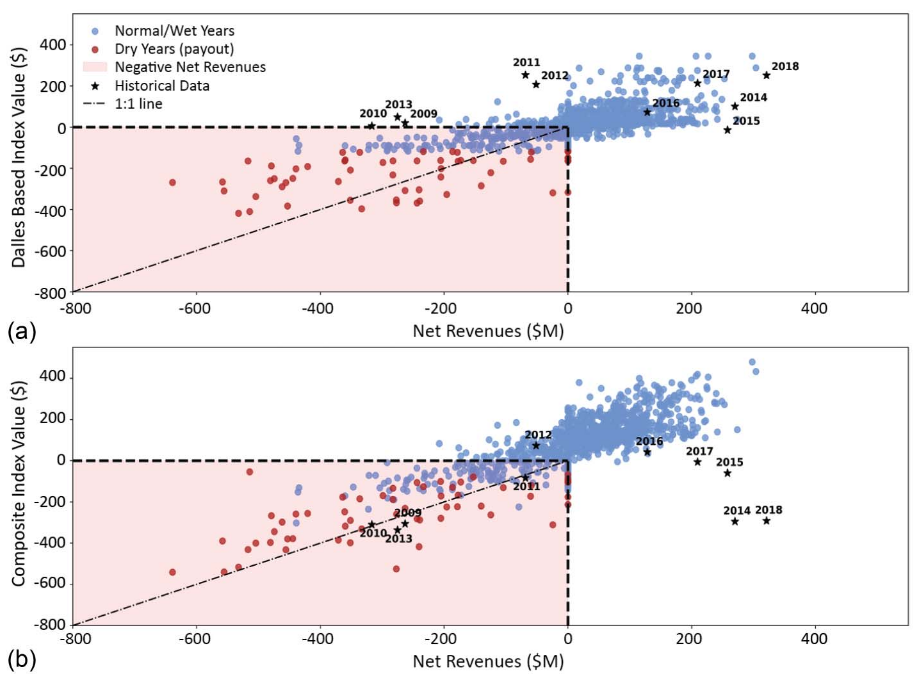

Fig. 4. Correlation of annual net revenues without risk management tools (i.e., reserves, CRAC surcharge) to: (a) an index based on annual Dalles streamflow; and (b) with a composite index that is a function of streamflow and electricity prices. Maroon points are years with Dalles streamflow at or below the fifth percentile of historical flow (years with payouts), while blue points are years with streamflow above the strike percentile. Star points are historical data from 2009 to 2018. The red shaded area highlights years with negative net revenues that are observed and predicted by the index.

to the index, a function of flow *and* Mid-C prices, when the strike is crossed and the index predicts negative net revenues, with a cap of $420 million (the ninety-ninth percentile of losses)

$$\text{Payout}(\$) = \begin{cases} 0 \leq -v \leq \text{cap}, & \text{if TDA} < S \text{ and } v < 0 \\ 0, & \text{otherwise} \end{cases} \quad (5)$$

The premium, paid by the BPA to the insurer, is calculated as the expected payout across simulated years plus a loading factor to account for the cost of transferring the risk of losses. One technique used to determine risk-adjusted premiums is the Wang transform (Wang 2002), an approach that incorporates consideration of the frequency and magnitude of payouts. The Wang transform is a function of a factor (λ), which is linked to the market price of risk and can be empirically derived. Transforming the distribution of payouts using λ is akin to weighing large, infrequent payouts more heavily when calculating the premium, at least in part accounting for the liquidity requirements placed on an insurer in order to make such payouts. As in previous work on weather derivatives and index-based instruments using the Wang transform (Baum et al. 2018; Foster et al. 2015; Hamilton et al. 2020), as well as the original research, this analysis sets λ to 0.25. Using the fifth percentile of annual TDA flow as the strike yields a premium of $15.7 million and an average payout of $12.5 million, with the loading contributing the remaining $3.2 million (roughly 25% of the average).

For the sake of comparison with the costs incurred by the US government, the cost of the premium is also assessed as an opportunity cost: the foregone earnings from putting funds toward the insurance premium as opposed to an investment which earns interest over time. This opportunity cost (O_c) is calculated as

$$O_c(\$) = \sum_{t=1}^{20} \frac{\text{prem} \cdot (1+i)^t - \text{prem}}{(1+d)^t} \quad (6)$$

where prem = the insurance premium ($); i = the annual interest rate for the alternative investment, which as in Eq. (1) is the 3.74% that could be earned in US treasuries; t = the time horizon for the investment, in years; and d = the discount rate, also consistent with Eq. (1).

Results

The following sections provide a comparison of the performance and cost of the risk management strategies. The first section evaluates the performance of each strategy. This provides the context for the second section, which quantifies the costs of each strategy and assesses to whom the costs accrue.

Effectiveness of Risk Management Tools

The objective of any risk management strategy is to reduce the risk of losses or negative net revenues. For the BPA, the risk of losses from hydrometeorological extremes is substantial; without its reserve fund, tariff surcharges, or line of credit, the BPA has a 5% chance of exceeding losses over $167 million (the 95% VaR), as seen in Fig. 5. The average loss above this 95% VaR, known as the contingent value at risk (CVAR) or expected shortfall, is far greater: −$309 million.

However, the BPA's current tools effectively reduce its risk. Each strategy improves the BPA's 95% VaR to $0, creating a bimodal

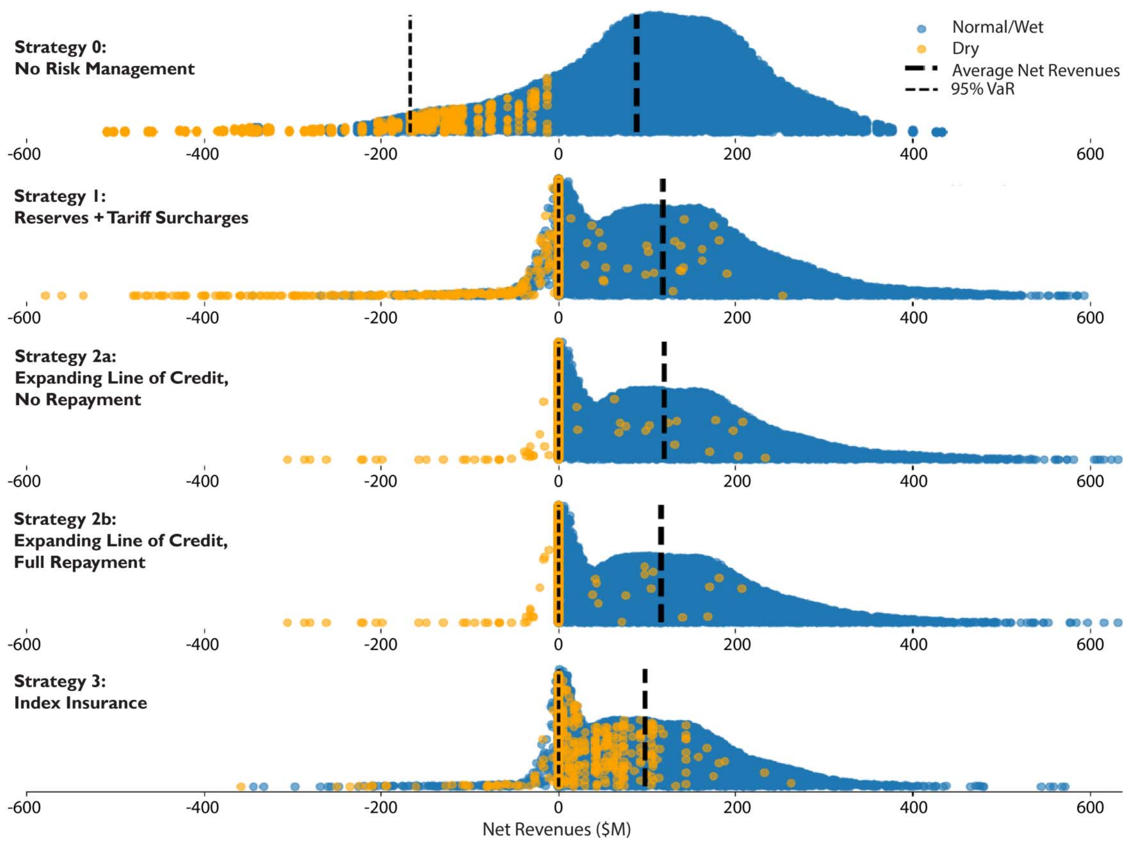

Fig. 5. Comparison of distributions of the BPA's annual net revenues across all strategies. Strategy 0 has no risk management tools. Strategy 1 only includes a reserve fund and tariff surcharges, with no line of credit. Strategy 2 includes an unlimited line of credit, with no (2a) or maximum (2b) repayment. Strategy 3 includes index insurance in lieu of the line of credit. The 95% VaR is fifth percentile of annual net revenues.

Table 2. Summary statistics for strategies

Strategy	Average net revenues ($M)	CVAR ($M)	Ensembles with tariff surcharges (%)	Mean tariff surcharge ($/MWh)
0. No risk management	88	−309	—	—
1. Reserves + surcharges	119	−25	97	0.48 (1.44%)
2. Unlimited line of credit	—	—	—	—
2a. No repayment	119	−1	67	0.24 (0.71%)
2b. Full repayment	117	−1	67	0.24 (0.71%)
3. Index insurance	98	−8	80	0.15 (0.46%)

Note: Only the unlimited line of credit strategies include a line of credit. The remaining strategies are independent of government support. Parentheses show mean tariff surcharges as a percentage of the average preference customer price.

distribution as losses are offset by use of the different risk management tools. Reserves and tariff surcharges alone reduce the CVAR to –$25 million (see Table 2), while the line of credit (Strategies 2a/2b) adds an additional layer of protection, improving the CVAR to –$1 million. Compared to the other tools, the line of credit is the most effective, as it is available in nearly all years and does not substantially raise costs for the BPA. Strategy 3, the use of index insurance in lieu of the line of credit, similarly improves the CVAR, to −$8 million, again, offsetting losses to varying degrees based on the index value. Unlike Strategies 1 and 2, however, the index introduces an element of basis risk and additional cost, whereby losses are not always sufficiently offset (or are offset to an excess), and years with positive, but low net revenues may newly become loss years because of the additional cost imposed by the premium. Strategies 1–3 all result in average annual net revenues ranging between $98 and $119 million, with the use of index insurance (Strategy 3) and payment of the associated premium resulting in the lowest average annual net revenues of the strategies. Use of the line of credit with no repayment (Strategy 2a) resulted in the highest average annual net revenues; overall, however, the differences in costs of risk management between strategies were relatively small. The more significant outcomes involve assessing to whom these costs accrue.

Costs of Risk Management

Results in the prior section indicate that the government-supported line of credit and index insurance effectively reduce the risk of losses when compared to no risk management (Strategy 0) and significantly reduce the CVAR even when compared to use of a reserve fund and tariff surcharges alone (Strategy 1). Given the comparable performance of the risk management strategies involving the line of credit and index insurance, it is useful to evaluate the costs of each strategy to the BPA's customers, the BPA (and longer-term its customers), and the US government (taxpayers).

First are the BPA's customers, who eventually absorb costs incurred by the BPA via longer-term rate changes but in the short-term are also faced with sudden tariff surcharges up to $5/MWh, which are triggered as a function of losses and remaining available risk management tools (i.e., availability of reserves and line of credit). When the BPA relies solely on reserves and tariff surcharges (Strategy 1), there are surcharges in 97% of the 590 ensembles with an average annual surcharge of $0.48/MWh (1.44% increase). Because surcharges are triggered in response to losses, the reduction in losses arising from use of either the line of credit (Strategy 2) or index insurance (Strategy 3) corresponds to a reduction in the percentage of ensembles with surcharges, from 97% to 67%–80%, along with a reduction in the magnitude of the surcharges, from $0.48/MWh to $0.15–$0.24/MWh (Table 2). Fig. 6(a) shows the summed value of the per MWh surcharges to highlight the aggregate costs for the BPA customers, an average PV between $6 and $20 million over 20 years depending on the strategy (see Table 3).

For the BPA, costs come in the form of line of credit repayments (debt service) for Strategy 2b and insurance premiums for Strategy 3. The average PV cost of line of credit payments in Strategy 2b is $37 million. It is additionally worth noting that this does not take into account any amortized payments beyond the 20-year time horizon. On average, $114 million of discounted payments remain outstanding between years 21 and 50, the remaining duration of the BPA's typical 50-year repayment schedule (BPA 2022). In Strategy 3, which involves index insurance, the BPA's cost is tied to payment of the insurance premium. Because payment of the premium uses funds that may have otherwise been invested elsewhere, the cost of the premium over the 20-year ensemble time horizon is calculated as an opportunity cost, the "missed" potential earnings from using funds to pay the premium as opposed to using them for a risk-free investment, such as treasury bonds. Were these funds invested, the annual $15.7 million premium would earn $272 million [Fig. 6(b)], a number that remains constant because of the fixed nature of the insurance premium.

Finally, the US government incurs most of the costs when the BPA relies on the line of credit, a combination of opportunity costs, foregone repayments, and interest rate subsidies. In the case of the government, the opportunity cost of the line of credit is assessed relative to the portion of the line that is held in liquid form for the BPA to access but which remains unused. Were the unused portion of the line invested by the government, earnings over 20 years would reach, on average, a present value of $443 million [Fig. 6(b)]. The second cost is foregone repayments, the difference between the scheduled line of credit payments that would lead to complete repayment, and the modeled repayment according to the two repayment strategies (Strategy 2a, no repayment; Strategy 2b, maximum repayment). Fig. 6(c) shows the average PV cost of foregone repayments, which ranges, on average, between $48 and $65 million. A final consideration is the subsidy the BPA receives: that is, the cost difference in borrowing at the below market interest rates offered by the government and market lenders. The average PV difference in the cost of repayment between the average interest rates for the BPA's market and treasury debt is approximately $22 million [see Fig. 6(d)].

These results address an important question: How does the cost-effectiveness of government intervention vis-à-vis an unlimited line of credit compare to an approach in which the BPA independently manages its risk? Table 3 summarizes the results for the immediate costs (tariff surcharges) to the BPA's customers, the costs borne by the BPA (and eventually passed to its customers), and the costs to the US government. Though use of any of the strategies presented here improves the 95% VaR to $0 and improves the CVAR substantially, the largest total PV costs accumulate in Strategies 2a and 2b with their reliance on the government-backed line of credit [an average of $539–$559 million as seen in Fig. 6(e)]. The cost of index insurance primarily accrues to the BPA as a fixed annual cost, with some contribution from the cost of surcharges. When this combined cost is discounted over 20 years, the ongoing $15.7 million annual premium and corresponding surcharges result in a PV cost of $278 million, far below the cost of strategies involving the line of credit.

The average PV cost, however, masks the large variability in costs over the range of ensembles. Tariff surcharges, which have a maximum cost because of their $5/MWh cap, account for between $0 and $66 million in costs, with the highest costs incurred in ensembles where tariff surcharges are triggered in the first years of the ensemble. Strategy 1 peaks at a cost of $66 million, while in Strategies 2 and 3 the value of tariff surcharges over the ensembles

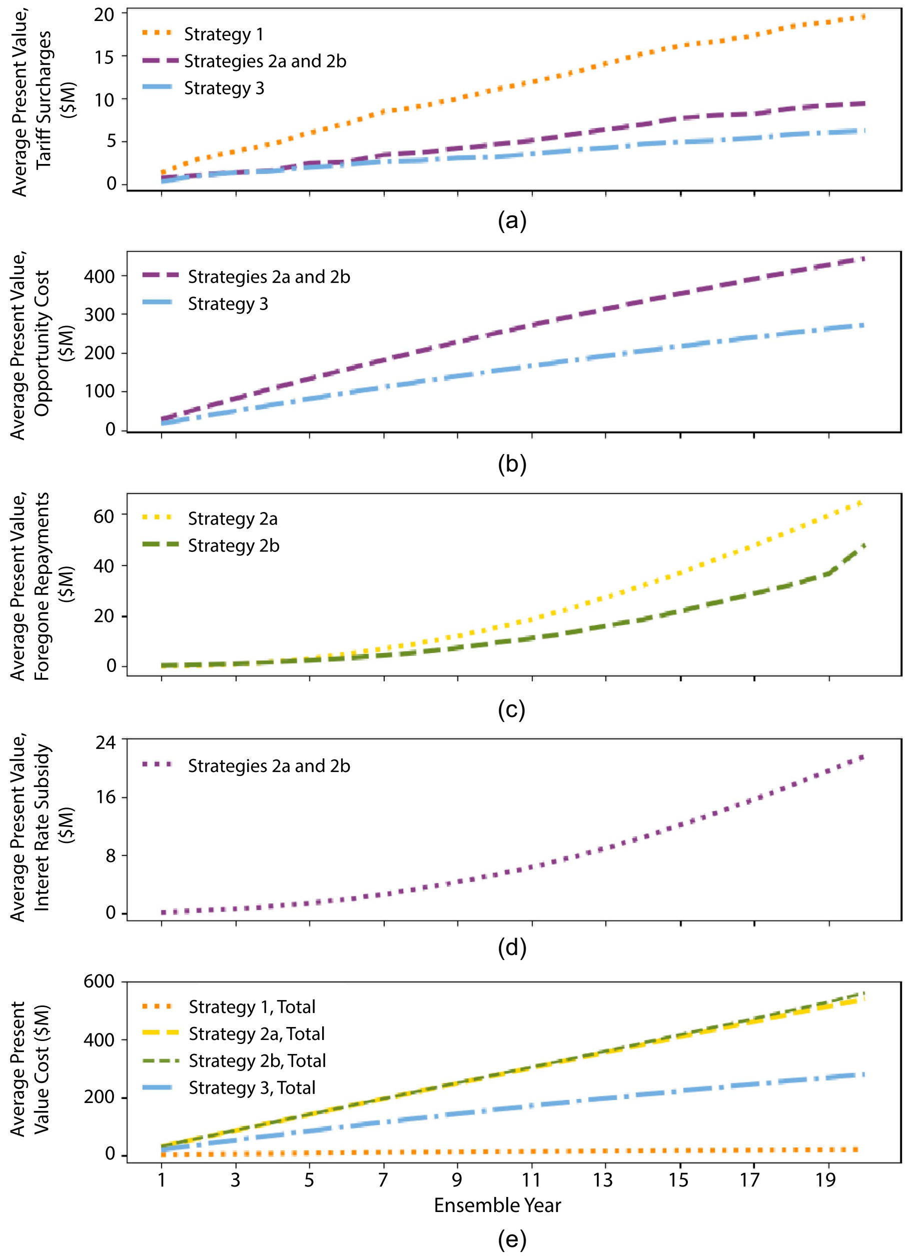

Fig. 6. Average present value cost of the line of credit as: (a) tariff surcharges; (b) opportunity cost; (c) foregone repayment; (d) interest rate subsidy; and (e) total cost.

Table 3. Average PV costs of risk management that accrue to the BPA's customers, annually to the BPA, and to the government, summed over 20 years

Strategy	Tariff surcharge costs (BPA customers) ($M)	Line of credit repayments and insurance premiums (BPA) ($M)	Foregone repayments and opportunity cost (government and taxpayers) ($M)	Average total cost ($M)
0. No risk management	—	—	—	—
1. Reserves + surcharges	20	0	0	20
2. Unlimited line of credit + reserves + surcharges	—	—	—	—
2a. No repayment	9	0	529	539
2b. Full repayment	9	37	512	559
3. Index insurance + reserves + surcharges	6	272	0	278

Note: Cost to the BPA include costs of line of credit repayments, and insurance premiums. Costs to the government are foregone repayments, the interest rate subsidy to the BPA, and the opportunity cost of holding liquid reserves for the BPA's use.

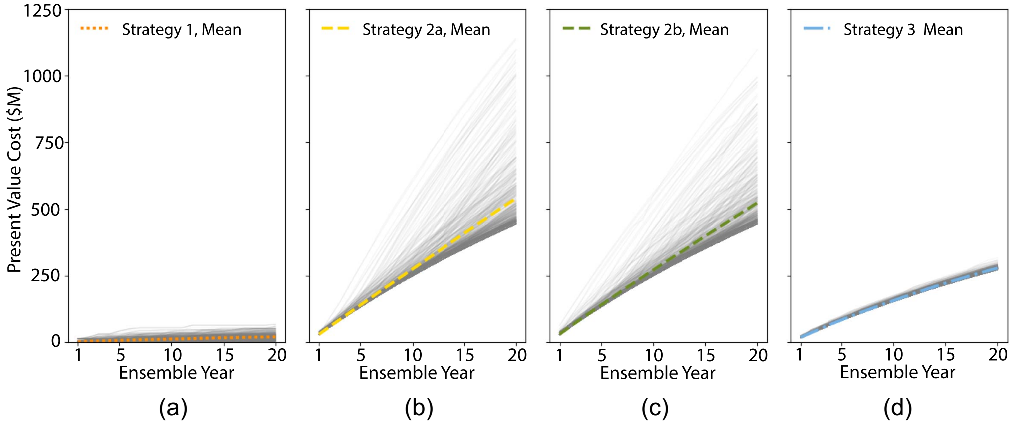

Fig. 7. Total PV costs for each ensemble (gray lines) and the average for (a) Strategy 1; (b) Strategy 2a; (c) Strategy 2b; and (d) Strategy 3.

reaches $43 and $45 million, respectively. The costs of foregone payments and implicit subsidies in Strategies 2a and 2b also vary. In ensembles in which the BPA relies more extensively on the line of credit, the implicit interest rate subsidy reaches over $197 million, foregone repayments reach $379/$526 million (Strategy 2a/Strategy 2b), and repayment from the BPA to the government reaches up to $415 million. Unlike the other tools, the cost of index insurance is constant because of the fixed premium.

The range of PV costs for each tool highlights an often-overlooked difference between risk management strategies: predictable costs. Strategy 1 has low costs with limited variability (Fig. 7), as it only includes reserves and tariff surcharges, the latter of which are capped, but it performs the most poorly in terms of risk management. Strategies 2a and 2b involve the line of credit with costs that vary greatly based on the number and severity of extreme events during each ensemble and corresponding use and repayment of the line of credit. This contrast is evident in simulation results: in a given sequence of years, the cost of Strategies 2a and 2b varies between 79%–82% and 196%–211% of the average PV cost ($539–559 million), from a minimum of $443 million to a maximum of over $1.1 billion [Figs. 7(b and c)]. This is far greater variability than Strategy 3, whose primary cost is the insurance premium, which is fixed for the duration of a contract irrespective of the number of loss events. The only variable costs of Strategy 3 are those due to tariff surcharges, triggered when insurance payouts are insufficient. The average PV cost of Strategy 3 is $278 million, a cost that varies from $272 to $316 million (98%–114% of average) (Fig. 7). These results are comparable regardless of discount rates used (see SM, Table S2 and Fig. S3).

Discussion

More than 10% of dams are in river basins facing high or very high risk of water scarcity (Opperman 2022). Such scarcity leaves hydropower suppliers exposed to significant financial risk, given the relationship between hydrometeorology and hydropower revenues (Senni et al. 2024). Though governments can subsidize the management of financial risk for hydropower suppliers, this may result in high and unpredictable costs that accrue to taxpayers and/or reduce government funds available for other priorities. As an alternative, suppliers may be able to manage their financial risk independently. This could include using cash reserves and tariff surcharges as well as an index-based risk transfer instrument, thereby leading the hydropower supplier (and its customers) to internalize the costs of risk management. Our results suggest that independent risk management can indeed be more cost-effective than a government-supported strategy in terms of managing the hydrometeorological risks even as more of these costs accrue to customers as opposed to a broader group of taxpayers. Results suggest another significant benefit to using an independent strategy involving index insurance: stable long-term costs.

In addition, the implications of each risk management strategy and the tools within it—reserves, tariff surcharges, the line of credit, and index insurance—raise questions about how a hydropower supplier like the BPA should weigh the costs of risk management: costs that accrue directly and immediately to customers (tariff surcharges); those accruing to the utility itself (debt service or premium payments); and those accruing to the government that supports the supplier (opportunity cost, foregone repayment, interest rate subsidies). Yet the trade-off between lower costs for customers and the utility versus for the government needs not be a zero-sum game. For example, the US government could subsidize an insurance premium that could help the BPA achieve its risk management goals while limiting and stabilizing the government's financial obligations to the BPA, passing off the risk of uncertain risk management costs to an insurer. In this scenario, the government could continue to provide the BPA (and its customers) with financial support equivalent to the average cost of the line of credit, but without large swings in costs. This may also help alleviate concerns that self-supported risk management would result in unaffordable prices for customers. Though political considerations play a role in decisions regarding government support, characterizing the costs of different tools and who bears these costs is a fundamental first step in designing a risk management policy. In addition, understanding the costs of these tools can help governments weigh potential indirect benefits from accepting certain aspects of financial risk. In the example of the BPA, one could argue that government support encourages maintenance of renewable energy resources and, more broadly, that it could incentivize investment in other renewable energy resources subject to similar hydrometeorological financial risk (e.g., wind). The impacts of such investments extend across the country, contributing to emissions reductions targets. Analysis of broader externalities may justify government spending.

The framework described in this research can extend to any hydropower supplier or government facing a choice as to how to manage its financial risk. These results are especially relevant in regions where hydropower shortfalls result in the need to activate expensive thermal generators, thereby increasing the costs of compensatory electricity purchases for suppliers who must meet firm electricity delivery contracts. In addition, climate change is generally expected

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to increase the frequency and severity of extreme events, including temperature and precipitation, with an expectation of increasingly correlated drought and heatwave events (Yin et al. 2022). In the PNW specifically, most climate change scenarios agree that climate change will lead to shifts in seasonal water availability from summer to winter and spring, as warmer temperatures lead to shifts from snow to rain and earlier snowmelt (Chegwidden et al. 2019). In terms of hydropower, these changes are expected to decrease water availability during the already dry season (June–September), while increasing temperatures will likely increase electricity demand (Hill et al. 2021; Zhou et al. 2018; Turner et al. 2019). Together, these changes would exacerbate generation shortfalls and therefore the need for (and costs of) compensatory electricity purchases. Indeed, research suggests that electricity shortfalls may more than double, with correspondingly large increases in electricity prices, particularly during late summer and early fall (Hill et al. 2021; Turner et al. 2019). These impacts will likely increase the BPA's hydrometeorological financial risk. As such, the costs of risk management may more frequently reach the upper bounds estimated here, or even surpass them, increasing the value of a strategy that has stable and predictable cost, as one with wildly varying costs does not manage financial risk effectively.

Alternative, and less expensive, strategies are likely even more important for hydropower generators and related power utilities in low and lower-middle income countries (LICs/LMICs). These entities may have smaller reserves to draw on than the BPA and in some cases may be entirely reliant on a national government with fewer resources, while tariff surcharges for customers may be more politically infeasible. For such countries, committing funds to a line of a credit may also have an outsize impact on budgetary spending, reducing funds available to invest in physical or social infrastructure and incurring high opportunity costs (Martinez-Diaz et al. 2019). The value of a line of credit or other borrowing, perhaps supplied by institutional backers such as the World Bank or Inter-American Development Bank, may thus be higher for LICs/LMICs than in the case of an entity such as the BPA and have more important implications at the national level. Additionally, a well-designed financial instrument could prevent the need to take on additional debt to offset losses under the most severe drought and heatwaves. Beyond simply considering opportunity cost, LICs/LMICs often have lower credit ratings, meaning that they are often only able to borrow at high interest rates (UN Inter-Agency Task Force on Financing for Development 2022). This makes access to debt and the ability to repay it more challenging than for high-income countries and utilities therein. Use of index insurance may reduce or even obviate the need for unexpected borrowing.

This study's results provide directions for future work. First, the framework for understanding the BPA's risk within a hydropower dominated electricity market could be extended to regions with a wider variety of renewable energy sources susceptible to weather variability (e.g., solar, wind). It may also be extended to any entity with government support that is exposed to hydrometeorological financial risk. Second, the likely increase in extreme weather events under climate change suggests the need for evaluating risk management strategies under different climate scenarios. Future research exploring both of these contexts may reveal even greater financial risk for entities like the BPA as well as a greater need for alternative risk management tools.

Conclusions

The objective of this work is to compare government- and self-supported hydrometeorological risk management strategies, including use of an index insurance contract. The BPA provides an excellent case study, given its status as a hydropower supplier with a mature risk management strategy and as one that also benefits from government support.

Results indicate that continued government intervention to manage the BPA's financial risk is, on average, more costly and subject to greater cost variability than letting the BPA manage its risk independently with index insurance layered on top of its existing reserves and tariff surcharges. If the index insurance contract designed in this work is used in lieu of the BPA's government-supported line of credit, the costs are not only lower in aggregate but accrue only to those that directly benefit from the risk management. The strategy involving index insurance in conjunction with reserves and tariff surcharges is also able to mitigate risk with more stable annual costs. At a time when many countries are seeking to increase their renewable energy portfolios, and developing countries in particular are expanding hydropower capacity, this analysis offers insights that may assist them in managing their hydrometeorological financial risk more cost-effectively.

Data Availability Statement

Models used are publicly available on GitHub: CAPOW (https://github.com/romulus97/CAPOW_PY36) and the BPA net revenue model/financial instrument design (https://github.com/UNC-Cofires/Comparing_Alts_Cuppari_etal). Historical data used are available on Zenodo (10.5281/zenodo.10565800) and are sourced from publicly available the BPA financial reports. Model runs from CAPOW are from Su et al. (2020a) and use data from Menne et al. (2012) and Sengupta et al. (2018). They are also available on Zenodo (10.5281/zenodo.14816798).

Reproducible Results

Yash Amonkar (University of North Carolina at Chapel Hill) downloaded all materials from GitHub and the data from Zenodo and ran the all_plots_final.py and sm_figures_tables_final.py scripts to reproduce the results shown in Figs. 3–7 and Tables 2 and 3 in the main text, as well as the results shown in Figs. S1 and S3 and Table S3 in the Supplemental Materials. The remaining figures and tables are conceptual or show starting conditions. The reproducibility reviewer also reproduced the results shown in all figures and tables.

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Author Contributions

Rosa I. Cuppari: Conceptualization; Data curation; Formal analysis; Investigation; Methodology; Validation; Visualization; Writing – original draft; Writing – review and editing. Simona Denaro: Conceptualization; Methodology; Writing – review and editing. Yufei Su: Methodology; Writing – review and editing. Jordan D. Kern: Methodology; Supervision; Writing – review and editing.

Gregory W. Characklis: Conceptualization; Funding acquisition; Methodology; Supervision; Writing – review and editing.

Supplemental Materials

Supplemental materials associated with this article are available online in the ASCE Library (www.ascelibrary.org).

References

Alizadeh, M. R., J. Adamowski, M. R. Nikoo, A. AghaKouchak, P. Dennison, and M. Sadegh. 2020. "A century of observations reveals increasing likelihood of continental-scale compound dry-hot extremes." *Sci. Adv.* 6 (39): eaaz4571. https://doi.org/10.1126/sciadv.aaz4571.

Apadula, F., A. Bassini, A. Elli, and S. Scapin. 2012. "Relationships between meteorological variables and monthly electricity demand." *Appl. Energy* 98 (Oct): 346–356. https://doi.org/10.1016/j.apenergy.2012.03.053.

Barnett, B. J., and O. Mahul. 2007. "Weather index insurance for agriculture and rural areas in lower-income countries." *Am. J. Agric. Econ.* 89 (5): 1241–1247. https://doi.org/10.1111/j.1467-8276.2007.01091.x.

Baum, R., G. W. Characklis, and M. L. Serre. 2018. "Effects of geographic diversification on risk pooling to mitigate drought-related financial losses for water utilities." *Water Resour. Res.* 54 (4): 2561–2579. https://doi.org/10.1002/2017WR021468.

Bloomberg News. 2022. "Hydropower production falls as dams from China to US hit by sweeping drought." Accessed December 26, 2022. https://www.bloomberg.com/news/features/2022-10-26/drought-from-china-to-us-hits-hydro-dams-slashing-the-top-clean-energy-source.

Board of Governors of the Federal Reserve System. 2022. "Market yield on US treasury securities at 30-year constant maturity, quoted on an investment basis." *Federal Reserve Economic Data*. Accessed March 14, 2023. https://fred.stlouisfed.org/series/DGS30.

Bodek, D., and J. Panger. 2021. *Bonneville power administration, Oregon; wholesale electric*. New York: S&P Global Ratings.

BPA (Bonneville Power Administration). 2017. *Power and transmission risk study.* Portland, OR: BPA.

BPA (Bonneville Power Administration). 2018. *Financial plan 2018*. Portland, OR: BPA.

BPA (Bonneville Power Administration). 2019. *2020 power rate schedules and general rate schedule provisions (FY 2020 2021)*. Portland, OR: BPA.

BPA (Bonneville Power Administration). 2022. "Shaping the future of clean energy." Accessed January 13, 2023. www.bpa.gov/-/media/Aep/finance/annual-techreports/ar2022.pdf.

Canale, L., G. Daci, H. Shijaku, and C. Ungerer. 2019. "Managing financial risks from hydropower." Accessed May 1, 2019. https://www.brookings.edu/blog/future-development/2019/05/01/managing-financial-risks-from-hydropower/.

Chegwidden, O. S., et al. 2019. "How do modeling decisions affect the spread among hydrologic climate change projections? Exploring a large ensemble of simulations across a diversity of hydroclimates." *Earth's Future* 7 (6): 623–637. https://doi.org/10.1029/2018EF001047.

CNA Insurance. 2016. *Risk transfer: A strategy to help protect your business*. Chicago: CNA Insurance.

Cregger, H., M. Atkins, E. Hoffmann, and L. Jones. 2021. *Water utilities prepared for current drought; climate risks will intensify threat over time*. New York: Moody's Investors Service.

Dakhlalla, A., S. Hughes, A. McManamon, E. Pytlak, T. R. Roth, and R. van der Zweep. 2020. *2020 level modified streamflow: 1928–2018*. Portland, OR: Bonneville Power Administration.

De Beaupuy, F., A. Nussbaum, and W. Horobin. 2022. "France to nationalize debt-laden EDF as energy crisis mounts." *Bloomberg*. Accessed May 10, 2023. https://www.bloomberg.com/news/articles/2022-07-06/french-premier-says-state-wants-to-own-100-of-edf.

DeFazio, P. A. 2021. "H.r.3684 infrastructure investment and jobs act." Accessed November 15, 2021. https://www.congress.gov/bill/117th-congress/house-bill/3684.

Denaro, S., R. I. Cuppari, J. D. Kern, Y. Su, and G. W. Characklis. 2022. "Assessing the Bonneville power administration's financial vulnerability to hydrologic variability." *J. Water Resour. Plann. Manage.* 148 (10): 05022006. https://doi.org/10.1061/(ASCE)WR.1943-5452.0001590.

Evans, S. 2019. "Blue marble & world food programme expand index insurance partnership." *Artemis.bm*. Accessed April 11, 2023. https://www.artemis.bm/news/blue-marble-world-food-programme-expand-index-insurance-partnership/#libraryItemId=14656478.

Fick, J. 2022. "Petrobras triples LNG imports in 2021 amid drought, pipeline work." *S&P Global Ratings*. Accessed December 27, 2022. https://www.spglobal.com/commodity-insights/en/news-research/latest-news/natural-gas/011322-petrobras-triples-lng-imports-in-2021-amid-drought-pipeline-work#libraryItemId=14131090.

Foster, B. T., J. D. Kern, and G. W. Characklis. 2015. "Mitigating hydrologic financial risk in hydropower generation using index-based financial instruments." *Water Resour. Econ.* 10 (Apr): 45–67. https://doi.org/10.1016/j.wre.2015.04.001.

Franke, A. 2022. "Droughts rattle Europe's hydropower market, intensifying energy crisis." *S&P Global*. Accessed September 7, 2022. https://www.spglobal.com/commodityinsights/en/market-insights/latest-news/electric-power/080522-droughts-rattle-europes-hydropower-market-intensifying-energy-crisis.

Greatrex, H., J. Hansen, S. Garvin, R. Diro, S. Blakeley, M. Le Guen, K. Rao, and D. Osgood. 2015. *Scaling up index insurance for smallholder farmers: Recent evidence and insights*. Rep. No. 14. Copenhagen, Denmark: CGIAR Research Program on Climate Change, Agriculture and Food Security.

Hamilton, A. L., G. W. Characklis, and P. M. Reed. 2020. "Managing financial risk trade-offs for hydropower generation using snowpack-based index contracts." *Water Resour. Res.* 56 (10): e2020WR027212. https://doi.org/10.1029/2020WR027212.

Hansrood, Z. 2022. "Record low Kariba dam levels see Zimbabwe, Zambia facing drastic power cuts." Accessed December 26, 2022. https://www.rfi.fr/en/africa/20221212-record-low-levels-at-kariba-dam-leave-zimbabwe-zambia-facing-drastic-power-cuts.

Hill, J., J. Kern, D. E. Rupp, N. Voisin, and G. Characklis. 2021. "The effects of climate change on interregional electricity market dynamics on the US West Coast." *Earth's Future* 9 (12): e2021EF002400. https://doi.org/10.1029/2021EF002400.

IIF (Index Insurance Forum). n.d. "Overview." *Global Index Insurance Facility.* Accessed April 11, 2023. https://www.indexinsuranceforum.org/overview.

Karakoyun, Y. 2024. "Determination of effective parameters for hydropower plants' energy generation: A case study." *Appl. Sci.* 14 (5): 2069. https://doi.org/10.3390/app14052069.

Kern, J. D., G. W. Characklis, and B. T. Foster. 2015. "Natural gas price uncertainty and the cost-effectiveness of hedging against low hydropower revenues caused by drought." *Water Resour. Res.* 51 (4): 2412–2427. https://doi.org/10.1002/2014WR016533.

Kim, C., and K. Krummenacker. 2019. "Moody's assigns Aa1 to energy Northwest (WA) Columbia generating station revenue bonds; affirms BPA and its supported debt obligations at Aa1; outlook revised to negative from stable." Accessed July 9, 2021. https://www.moodys.com/research/Moodys-assigns-Aa1-to-Energy-Northwest-WA-Columbia-Generating-Station-Rating-Action–PR_905774475.

Kim, C., and K. Krummenacker. 2020. "Moody's downgrades BPA (OR) to Aa2 from Aa1; assigns Aa2 rating to Morrow (port of) OR's transmission revenue bonds; outlook is stable." Accessed March 1, 2021. https://www.moodys.com/research/Moodys-downgrades-BPA-OR-to-Aa2-from-Aa1-assigns-Aa2-Rating-Action–PR_906313039.

Kim, C., and A. Sabatelle. 2012. "Moody's assigns Aa1 rating to BPA backed lease revenue bonds issued by Port of Morrow (OR). Rating outlook is stable." Accessed June 1, 2022. https://www.moodys.com/research/Moodys-assigns-Aa1-rating-to-BPA-backed-lease-revenue-bonds-Rating-Action–PR_250401.

Kim, C., and A. Sabatelle. 2022. "Moody's affirms Bonneville Power Administration's (OR) Aa2 issuer rating; outlook is revised to positive from stable." Accessed May 31, 2022. https://www.moodys.com/research/Moodys-affirms-Bonneville-Power-Administrations-OR-Aa2-issuer-rating-outlook-Rating-Action–PR_907640117?cy=kor&lang=ko.

Kim, C. J., J. A. Ponte, and A. J. Sabatelle. 2023. "Credit update following outlook change to negative and affirmation of Aa1 issuer rating." Accessed December 16, 2023. https://www.bpa.gov/-/media/Aep/finance/investor-relations/moodys.pdf.

Kim, C. J., A. J. Sabatelle, and C. M. Hu. 2016. "New issue-Moody's assigns Aa1 to energy Northwest's (WA) CGS, Project 1 and Project 3 revenue bonds. Rating outlooks are stable." Accessed July 10, 2023. https://www.energy-northwest.com/whoweare/finance/Documents/Investor%20Relations/BPA-ENW%20new%20sale%20march%202016-final.pdf.

Kleiman, R. M., G. W. Characklis, and J. D. Kern. 2022. "Managing weather-and market price-related financial risks in algal biofuel production." *Renewable Energy* 200 (Nov): 111–124. https://doi.org/10.1016/j.renene.2022.09.104.

Klein, M. 1997. "The risk premium for evaluating public projects." *Oxford Rev. Econ. Policy* 13 (4): 29–42. https://doi.org/10.1093/oxrep/13.4.29.

Kron, W., P. Löw, and Z. W. Kundzewicz. 2019. "Changes in risk of extreme weather events in Europe." *Environ. Sci. Policy* 100 (Oct): 74–83. https://doi.org/10.1016/j.envsci.2019.06.007.

Leong, M., W. Yu, and P. Chen. 2022. "Fitch affirms China three gorges and Yangtze power at 'a+'; outlook stable." *Fitch Ratings*. Accessed December 26, 2022. https://www.fitchratings.com/research/corporate-finance/fitch-affirms-china-three-gorges-yangtze-power-at-a-outlook-stable-23-09-2022.

Liao, S.-C., S.-C. Chang, and T.-C. Cheng. 2021. "Managing the volatility risk of renewable energy: Index insurance for offshore wind farms in Taiwan." *Sustainability* 13 (16): 8985. https://doi.org/10.3390/su13168985.

Lowrey, D. 2017. "Utility stock performance in a rising interest rate environment." *S&P Global Market Intelligence*. Accessed November 24, 2023. https://www.spglobal.com/marketintelligence/en/news-insights/research/utility-stock-performance-in-a-rising-interest-rate-environment.

Lucas, D. 2014. "Evaluating the cost of government credit support: The OECD context." *Econ. Policy* 29 (79): 553–597. https://doi.org/10.1111/1468-0327.12034.

Martinez-Diaz, L., L. Sidner, and J. McClamrock. 2019. *The future of disaster risk pooling for developing countries: Where do we go from here?* Washington, DC: World Resources Institute.

Matsumoto, T., and Y. Yamada. 2021. "Customized yet standardized temperature derivatives: A non-parametric approach with suitable basis selection for ensuring robustness." *Energies* 14 (11): 3351. https://doi.org/10.3390/en14113351.

Mazdiyasni, O., and A. AghaKouchak. 2015. "Substantial increase in concurrent droughts and heatwaves in the United States." *Proc. Natl. Acad. Sci. USA* 112 (37): 11484–11489. https://doi.org/10.1073/pnas.1422945112.

Mechler, R., S. Hochrainer, G. Pflug, A. Lotsch, and K. Williges. 2010. "*Assessing the financial vulnerability to climate-related natural hazards*." Accessed December 18, 2023. http://hdl.handle.net/10986/9048.

Menne, M. J., et al. 2012. *Global historical climatology network—Daily (GHCN-daily), version 3*. Asheville, NC: NOAA National Centers for Environmental Information.

Meyer, E. S., G. W. Characklis, and C. Brown. 2017. "Evaluating financial risk management strategies under climate change for hydropower producers on the Great Lakes." *Water Resour. Res.* 53 (3): 2114–2132. https://doi.org/10.1002/2016WR019889.

Meyer, E. S., G. W. Characklis, C. Brown, and P. Moody. 2016. "Hedging the financial risk from water scarcity for great lakes shipping." *Water Resour. Res.* 52 (1): 227–245. https://doi.org/10.1002/2015WR017855.

Meyer, E. S., B. T. Foster, G. W. Characklis, C. Brown, and A. J. Yates. 2020. "Integrating physical and financial approaches to manage environmental financial risk on the great lakes." *Water Resour. Res.* 56 (5): e2019WR024853. https://doi.org/10.1029/2019WR024853.

NWPCC (Northwest Power and Conservation Council). 2019. "Pacific Northwest hydropower for the 21st century power grid." Accessed April 5, 2020. https://www.nwcouncil.org/energy/energy-topics/hydropower.

OMB (Office of Management and Budget). 2022. "OMB circular No. A-94 Appendix C." Accessed June 22, 2023. https://www.whitehouse.gov/wp-content/uploads/2023/12/CircularA-94AppendixC.pdf.

Opperman, J. 2022. "Hydropower and water scarcity: The growing climate risks of a climate solution." Accessed December 26, 2022. https://www.forbes.com/sites/jeffopperman/2022/12/16/hydropower-and-water-scarcity-the-growing-climate-risks-of-a-climate-solution/?sh=2ee5c6b27258.

Rack, Y. 2020. "Climate impact on water supplies puts squeeze on hydropower." Accessed April 6, 2022. https://www.spglobal.com/marketintelligence/en/news-insights/latest-news-headlines/climate-impact-on-water-supplies-puts-squeeze-on-hydropower-59293719.

Sengupta, M., Y. Xie, A. Lopez, A. Habte, G. Maclaurin, and J. Shelby. 2018. "The national solar radiation data base (NSRDB)." *Renewable Sustainable Energy Rev.* 89 (Jun): 51–60. https://doi.org/10.1016/j.rser.2018.03.003.

Senni, C. C., S. Goel, and A. von Jagow. 2024. "Economic and financial consequences of water risks: The case of hydropower." *Ecol. Econ.* 218 (Apr): 108048. https://doi.org/10.1016/j.ecolecon.2023.108048.

Soares, J. M., and L. L. Hess. 2014. "Announcement: Protracted drought to continue pressure on Brazilian hydro generators' cash flows." Accessed August 29, 2022. https://www.moodys.com/research/Moodys-Protracted-drought-to-continue-pressure-on-Brazilian-hydro-generators–PR_309538?WT.mc_id=AMWWFob29fRmluYW5jZTQyX1NCX1JhdGluZyBOZXdzX0FsbF9Fbmc%3D20140930_PR_309538.

Su, Y., J. D. Kern, S. Denaro, J. Hill, P. Reed, Y. Sun, J. Cohen, and G. W. Characklis. 2020a. "An open source model for quantifying risks in bulk electric power systems from spatially and temporally correlated hydrometeorological processes." *Environ. Modell. Software* 126 (Apr): 104667. https://doi.org/10.1016/j.envsoft.2020.104667.

Su, Y., J. D. Kern, P. M. Reed, and G. W. Characklis. 2020b. "Compound hydrometeorological extremes across multiple timescales drive volatility in California electricity market prices and emissions." *Appl. Energy* 276 (Oct): 115541. https://doi.org/10.1016/j.apenergy.2020.115541.

Swiss Re. 2018. *Closing the gap drought threat to Uruguay's reliance on hydropower*. Zurich, Switzerland: Swiss Re.

Swiss Re. 2022. "Flow: Parametric water-level insurance." *Swiss Re Corporate Solutions*. Accessed April 11, 2023. https://corporatesolutions.swissre.com/dam/jcr:3b534611-27be-4fc1-b2dd-1e570a9d882a/FLOW-parametric-water-level-insurance.pdf.

The World Bank. 2017. *Sovereign climate and disaster risk pooling: World Bank technical contribution to the G20*. Washington, DC: The World Bank.

The World Bank. 2013. "The world bank partners with Uruguay to execute largest public weather and oil price insurance transaction." Accessed July 9, 2021. https://www.worldbank.org/en/news/press-release/2013/12/19/world-bank-uruguay-public-weather-oil-price-insurance-transaction.

Thomson Reuters. 2023. "Henry hub natural gas spot price (dollars per million btu)." *US EIA*. Accessed October 26, 2023. https://www.eia.gov/dnav/ng/hist/rngwhhdm.htm.

Turner, S. W. D., N. Voisin, J. Fazio, D. Hua, and M. Jourabchi. 2019. "Compound climate events transform electrical power shortfall risk in the Pacific Northwest." *Nat. Commun.* 10 (1): 8. https://doi.org/10.1038/s41467-018-07894-4.

UN Inter-Agency Task Force on Financing for Development. 2022. *Financing for sustainable development techreport 2022*. New York: United Nations.

USDA (US Dept. of Agriculture Risk Management Agency). n.d. "Rainfall index." Accessed April 11, 2023. https://www.rma.usda.gov/Policy-and-Procedure/Insurance-Plans/Rainfall-Index.

US EIA (US Energy Information Administration). 2023. "Wholesale electricity and natural gas market data." Accessed October 6, 2021. https://www.eia.gov/electricity/wholesale/.

Wang, S. S. 2002. "A universal framework for pricing financial and insurance risks." *ASTIN Bull.* 32 (2): 213–234. https://doi.org/10.2143/AST.32.2.1027.

WMO (World Meteorological Organization). 2022. *2022 state of climate services: Energy*. Geneva: WMO.

Yao, J. 2021. *Electricity consumption and temperature: Evidence from satellite data*. Rep. No. 2021/022. Washington, DC: International Monetary Fund.

Yin, J., L. Slater, L. Gu, Z. Liao, S. Guo, and P. Gentine. 2022. "Global increases in lethal compound heat stress: Hydrological drought hazards under climate change." *Geophys. Res. Lett.* 49 (18): e2022GL100880. https://doi.org/10.1029/2022GL100880.

Zeff, H. B., and G. W. Characklis. 2013. "Managing water utility financial risks through third-party index insurance contracts." *Water Resour. Res.* 49 (8): 4939–4951. https://doi.org/10.1002/wrcr.20364.

Zhou, T., N. Voisin, and T. Fu. 2018. "Non-stationary hydropower generation projections constrained by environmental and electricity grid operations over the Western United States." *Environ. Res. Lett.* 13 (7): 074035. https://doi.org/10.1088/1748-9326/aad19f.

J. Water Resour. Plann. Manage., 2025, 151(11): 04025062